

Sam Jackson

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EDUCATION

Master of Science in Computer Science

Anticipated May 2028

Cybersecurity Graduate Certificate

University of Arkansas, Fayetteville, AR

Bachelor of Science in Computer Science

August 2022 - May 2026

Double Minor in Mathematics and STEM Education | Concentration in Cybersecurity

University of Arkansas, Fayetteville, AR | GPA 3.534

- Alumni of Phi Sigma Rho Engineering Sorority, serving as Media Officer.
- Founded RazorBook Club, a university-recognized student organization, and served as Communications Chair.

WORK AND LEADERSHIP EXPERIENCE

Graduate Research Assistant

May 2026 - Present

Resilient Information and Operational Technologies Lab | Univ. of Arkansas

- Built working knowledge of semiconductor fabrication, including epitaxial growth, photolithography, and ion implantation.
- Developing a thesis proposal that combines physics-informed machine learning with TCAD simulation to optimize SiC epitaxial growth parameters through defect detection at the Multi-User Silicon Carbide (MUSiC) facility.
- Design integrated circuit layouts in KLayout, applying device physics to extend prior TCAD simulation work.
- Participate in design rule check (DRC) reviews for integrated circuit layouts.

Research Experience for Undergraduates

May 2026 - July 2026

University of Arkansas Power Group | Univ. of Arkansas

- Investigated the effects of process parameters on aluminum ion implantation in 4H-SiC for power semiconductor fabrication.
- Developed proficiency in Synopsys Sentaurus TCAD and Monte Carlo (BCA) simulation for semiconductor process modeling.
- Completed certification training in Ignition SCADA, Canary Historian, PyTorch, and NVIDIA Jetson platforms.

Research Assistant

December 2025 - May 2026

Resilient Information and Operational Technologies Lab | Univ. of Arkansas

- Developing hands-on hardware skills including soldering, PLC programming, wiring diagram creation, and FPGA-based systems.
- Contributing to research initiatives supporting the National Center for Reliable Electric Power Transmission.
- Supporting the Cybersecurity Consortium for Innovation project by helping design cybersecurity training modules and classroom exercises to teach foundational security concepts.

Software Engineering Intern

May 2025 - May 2026

Union Pacific Railroad | Omaha, NE and Remote

- Led migration of a legacy Ricola application to an Angular 19, mobile-responsive interface, supporting 8K+ field users and 1.5M monthly requests.
- Implemented Figma designs into scalable, production-ready frontend components.
- Optimized backend performance with Java Spring Boot via asynchronous and batched API calls, scheduled notification jobs, and GeoJSON-based database updates.
- Collaborated in an Agile environment with daily standups, sprint planning, and Git-based version control.

PROJECTS, EVENTS, PERSONAL PROJECTS

HogHacks Hackathon

April 2025

Association for Computing Machinery | Univ. of Arkansas

- Built a web-based legal case exploration platform with an interactive chat simulation in 24 hours using JavaScript, HTML, and CSS.
- Tied for 2nd place and awarded Most Creative Project.